Geometry
Unit 7
Lesson 6 Practice

Name_Answer Key________
Date___________________
Period__________________

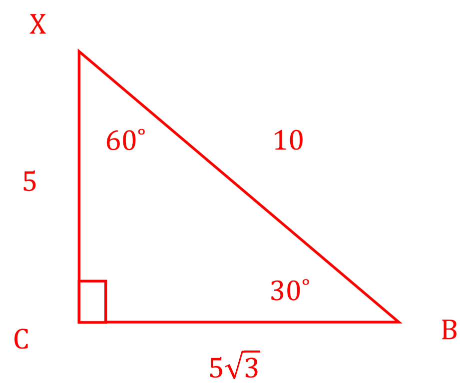
$\triangle BXC$ is a right triangle where $m\angle XCB = 90°$, $m\angle CXB = 60°$, and $XC = 5$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
1.	$\overline{CB}$	$5\sqrt{3}$
2.	$\overline{XB}$	10

Hypotenuse is double the short leg (5),

so hypotenuse is 10.

Long leg is short leg $\cdot\sqrt{3}$ $(5\sqrt{3})$.

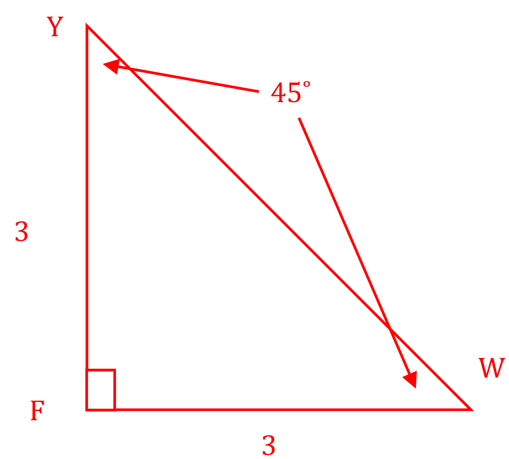


$\triangle YFW$ is a right triangle where $m\angle YFW = 90°$, $m\angle FYW = 45°$, and $FY = 3$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
3.	$\overline{WF}$	3
4.	$\overline{YW}$	$3\sqrt{2}$

Isosceles so $FW = 3$, also hypotenuse

is leg$\cdot\sqrt{2}$ so $3\sqrt{2}$

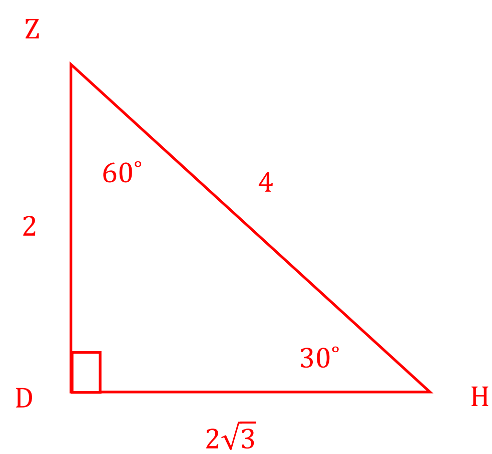


$\triangle DZH$ is a right triangle where $m\angle ZDH = 90°$, $m\angle DHZ = 30°$, and $ZH = 4$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
5.	$\overline{ZD}$	2
6.	$\overline{DH}$	$2\sqrt{3}$

Short leg is half the hypotenuse

Long leg is short leg $\cdot\sqrt{3}$ $(2\cdot\sqrt{3})$

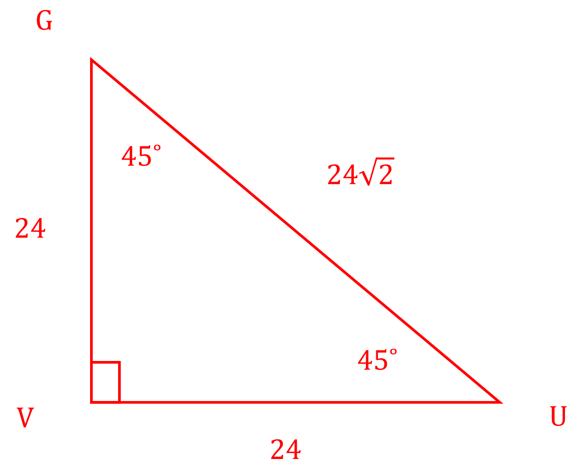


$\triangle GVU$ is a right triangle where $m\angle GVU = 90°$, $m\angle UGV = 45°$, and $VG = 24$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
7.	$\overline{VU}$	24
8.	$\overline{GU}$	$24\sqrt{2}$

$VU = 24$ since both legs are congruent

Hypotenuse is leg $\cdot \sqrt{2}$



$\triangle JTS$ is a right triangle where $m\angle TSJ = 90°$, $m\angle JTS = 60°$, and $TJ = 12$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
9.	$\overline{TS}$	6
10.	$\overline{SJ}$	$6\sqrt{3}$

Short leg $\overline{TS}$ is half the hypotenuse.

Long leg is short leg $\cdot \sqrt{3}$ $(6\sqrt{3})$.